

AdminLens — Generative AI Developer Specification

Data Ingestion, API Mappings & Technical Architecture Guide

1. GOOGLE WORKSPACE API DATA INGESTION

Data is extracted from the **Google Admin SDK Reports API** (`reports_v1`). Authentication requires a Google Cloud Service Account with Domain-Wide Delegation granted the OAuth scope: `https://www.googleapis.com/auth/admin.reports.audit.readonly` .

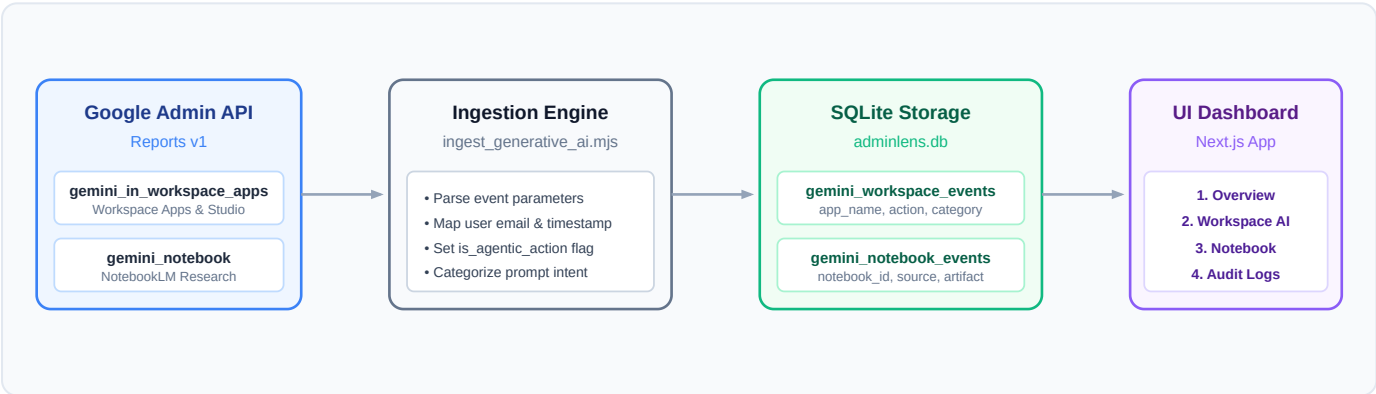
Endpoint & Calling Method

```
GET https://admin.googleapis.com/admin/reports/v1/activity/users/{userKey}/applications/{applicationName}
```

Call with `userKey: 'all'` and `maxResults: 100` . Loop through `nextPageToken` until all records are retrieved. AdminLens ingests two distinct `applicationName` streams:

Application Stream	Description & Purpose	Monitored Activities / Event Names
<code>gemini_in_workspace_apps</code>	Tracks prompt activity and automated AI features across core Google Workspace apps.	Gemini App, Google Docs writing, Gmail unread triage, Meet studio sound, Slides image creation, Workspace Studio automated flows.
<code>gemini_notebook</code>	Tracks grounded research notebooks, reference sources, and generated studio content.	Create notebook, add source (PDF/Docs/URLs), Audio Overview podcast generation, research note creation, visibility updates.

2. END-TO-END PIPELINE & DATABASE MAPPING ARCHITECTURE



SQLite Schema Structure

```
-- Workspace Apps: gemini_workspace_events
CREATE TABLE gemini_workspace_events (
  id TEXT PRIMARY KEY, event_time TEXT NOT NULL, user_email TEXT NOT NULL,
  app_name TEXT, action TEXT, event_category TEXT, is_agentic_action INTEGER DEFAULT 0, raw_json TEXT
);
-- Gemini Notebook: gemini_notebook_events
CREATE TABLE gemini_notebook_events (
  id TEXT PRIMARY KEY, event_time TEXT NOT NULL, user_email TEXT NOT NULL, event_name TEXT NOT NULL,
  notebook_id TEXT, notebook_title TEXT, source_name TEXT, studio_artifact_name TEXT, raw_json TEXT
);
```

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UI Views, Component Layouts & Table Breakdown

3. MODULE VIEWS: TABLES & GRAPHS SPECIFICATION

The module is located at route `/generative-ai?tab={tabName}` and contains four operational views:

● **Tab 1: Overview (?tab=overview)**

Page Header: Generative AI Overview

Top KPI Cards (4): Total Activity Events, Active Users, Autonomous Actions (Studio flows), Gemini Notebook Projects.

Visual Graphs (2x2 Balanced Grid):

- **Generative AI Activity:** Timeline line/area chart with dual toggles (Workspace vs Notebook / User vs Agentic).
- **AI Activity by Application:** Doughnut chart with center badge displaying total events across apps.
- **AI Capabilities & Tasks:** Bar chart switchable between Core Pillars and Top Actions.
- **How AI Is Used:** Stacked bar showing engagement split (Conversations vs Content vs Passive).

Table: User Activity — Account leaderboard with real-time email search, column sorting, and pagination (10/25/50/100).

● **Tab 2: Gemini in Workspace (?tab=apps)**

Page Header: Gemini in Workspace (with Google Gemini 2025 icon)

Top KPI Cards (3): Total Events in Workspace, Execution Mode (User vs Automated flow ratio), Top Workspace App.

Tables (2):

- **Application Usage Table:** Shows integrated apps (Docs, Gmail, Slides, Meet, Studio), event counts, visual "Share of Activity" progress bar, and mode pill.
- **Application Features Table:** Lists specific AI actions (e.g. story generation, proactive summary), event count, visual share bar, category, and execution mode badge. Supports sorting.

● **Tab 3: Gemini Notebook (?tab=notebooks)**

Page Header: Gemini Notebook (with NotebookLM icon)

Top KPI Cards (3): Total Notebooks (shared vs private count), Ingested Sources (reference docs), Generated Studio Artifacts.

Tables (2):

- **Notebook Projects Table:** Title, owner email, creation date, visibility tag (Shared vs Private), and ingested sources count.
- **Ingested Reference Sources Table:** Source title, document type pill (PDF, Google Doc, Web URL), parent notebook, and ingestion date.

● **Tab 4: Audit Logs (?tab=audit)**

Page Header: Generative AI Audit Logs

Purpose: Granular, compliance-grade event viewer merging both workspace and notebook activity streams.

Controls & Filters:

- Application dropdown filter (Gmail, Meet, Docs, Studio, etc.)
- Actor execution mode filter (All, User-initiated, Agentic flow)
- Engagement filter (Active prompt vs Passive background)
- Free-text keyword search filter across user and action

Inspection Modal: Clicking any event row opens a drawer/modal displaying exact timestamp, user ID, client ID, category code, and complete raw JSON payload.

4. KEY RULES FOR DEVELOPER IMPLEMENTATION

- **Permanent Overview Graphs:** Overview charts must remain permanently rendered with no close/remove buttons.
- **Equal Row Heights:** All charts on the Overview page use `flex flex-col h-full` and `items-stretch` to ensure left and right cards line up evenly.
- **Client Performance:** Heavy event logs are pre-aggregated into JSON caches (`generative_ai_data.json`) during daemon ingestion so client page loads remain sub-second.